Lesson 17:

Maintenance



IGSS

Interactive Graphical
SCADA System

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Contents of this lesson



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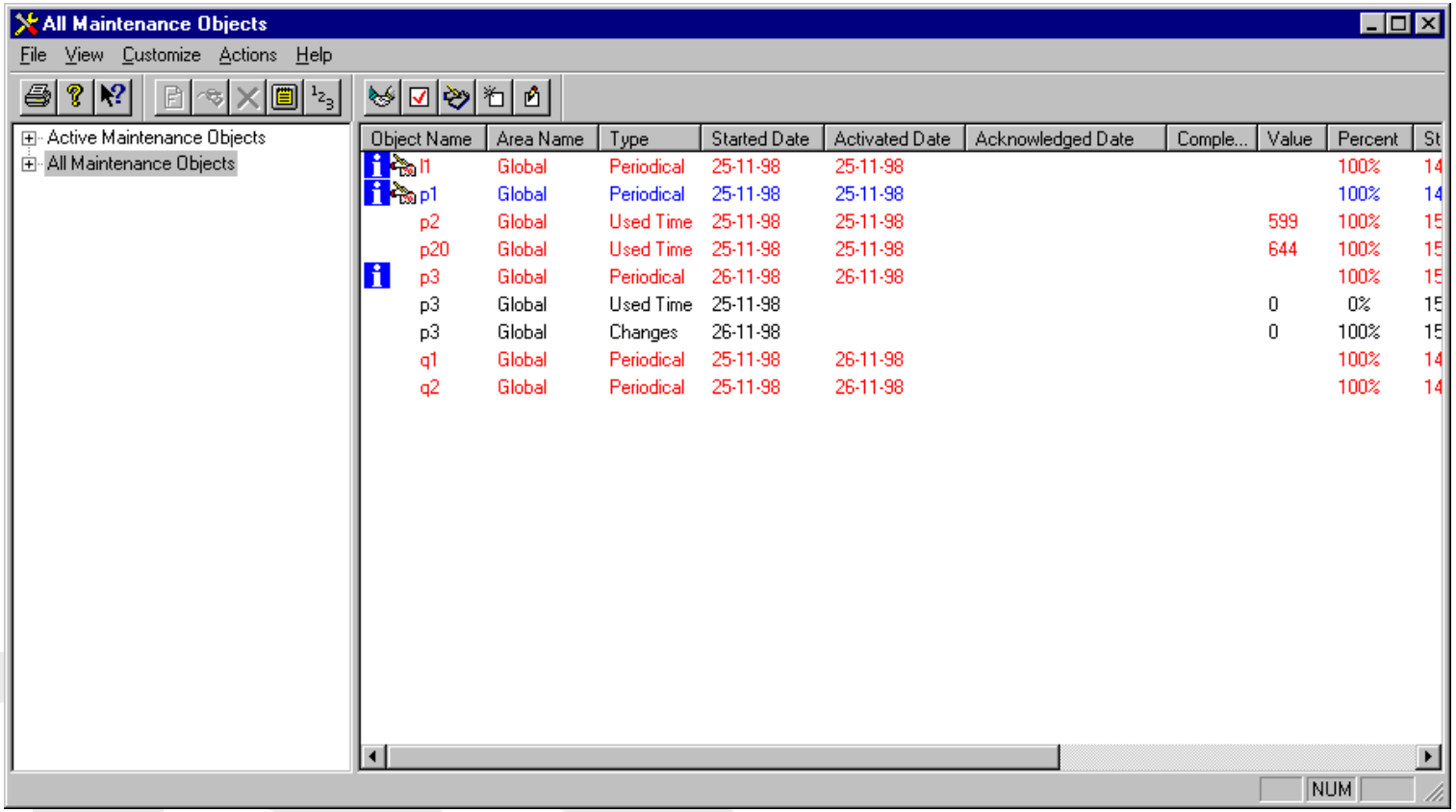
The Maintenance module



- Allows users to create maintenance jobs to ensure timely maintenance on the process components
- Show maintenance alarms to ensure that the service personnel will perform the maintenance job on time
- The Maintenance module is included in the Standard version of IGSS

The Maintenance List

- Very similar to Alarm List
- New fields: Type and Percent



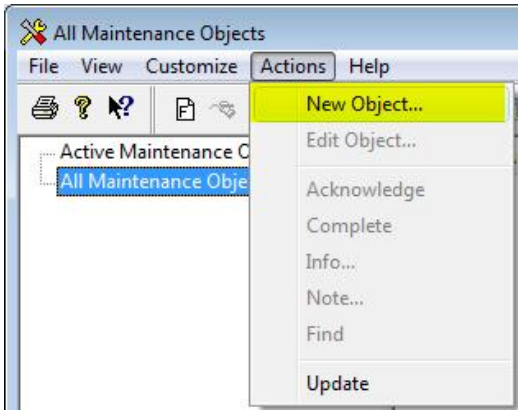
Maintenance types



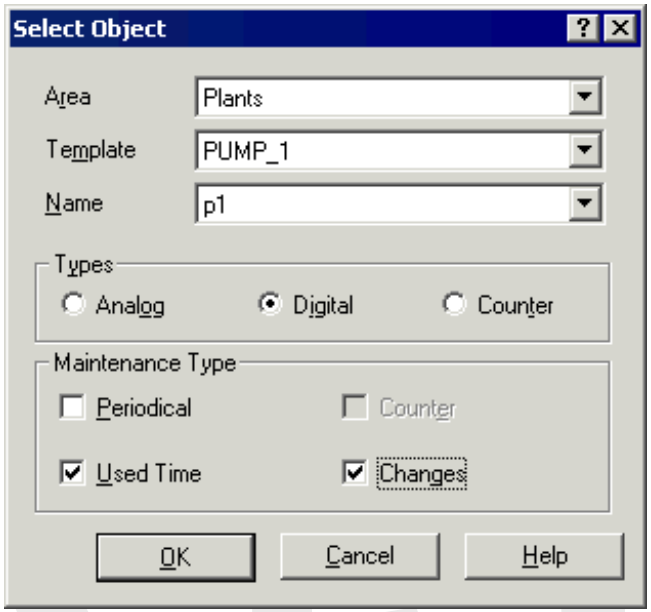
Type	Description
Periodical	Fixed maintenance interval
Counter	Fixed value or counter
Used Time	Hours of operation (digital only)
Changes	Number of state changes (digital only)

1. You can combine the types.
   Example: Periodical, Used Time and Changes.
2. You can have more than one instance of the same type for one object.
   Example: 500 hours and 1500 hours periodical jobs for an analog object.

Creating a Maintenance job



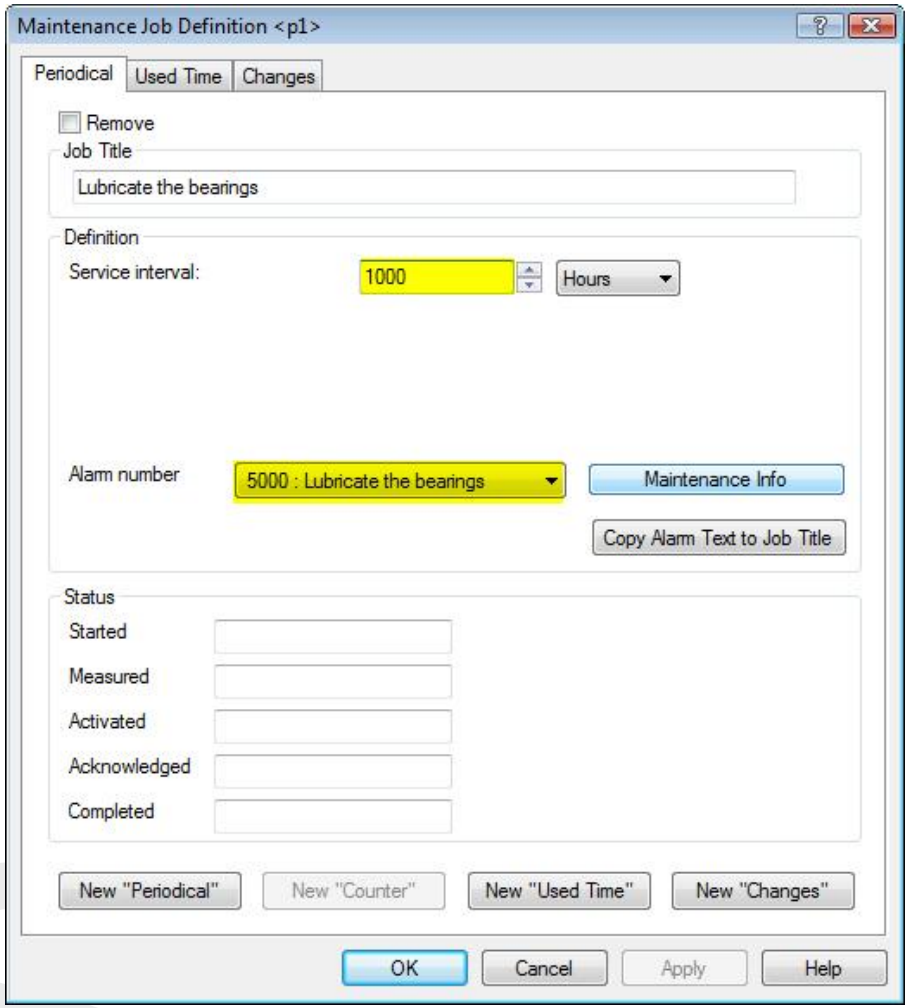
1. In the **Actions** menu, select **New Object**.



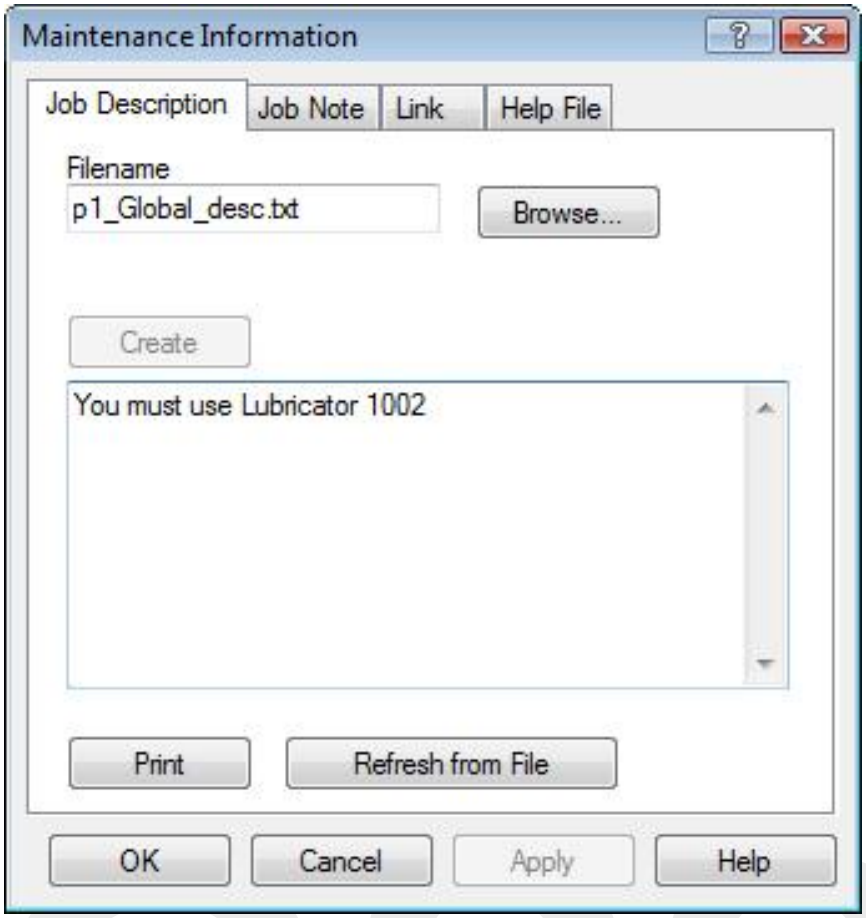
2. Select the object type.
3. Select the object name.
4. Select the maintenance type(s).

Attach alarm and instructions

1. Specify service interval.
2. Attach alarm text to show maintenance job in the Alarm List.



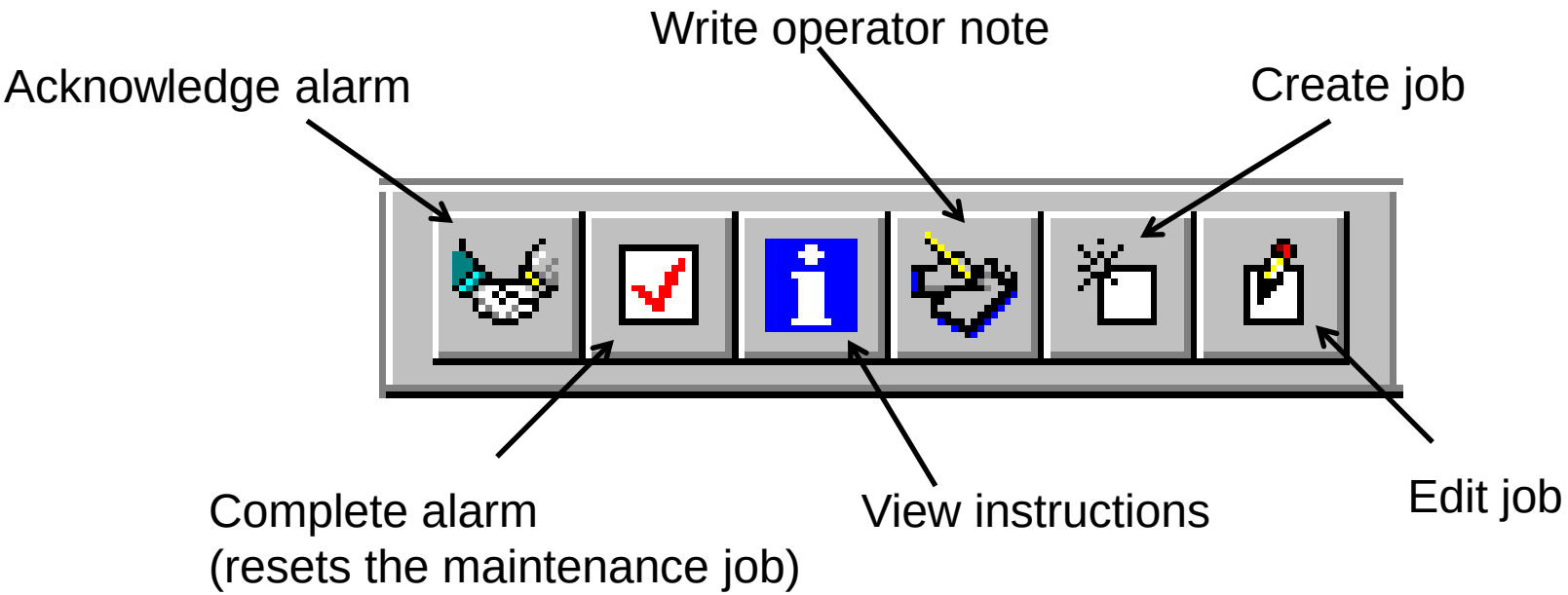
Instructions as text or file link



- **Job Description tab**
  Write instructions in text file
- **Link tab**
  Link to an electronic manual or video
- **Help File tab**
  Link to help topic

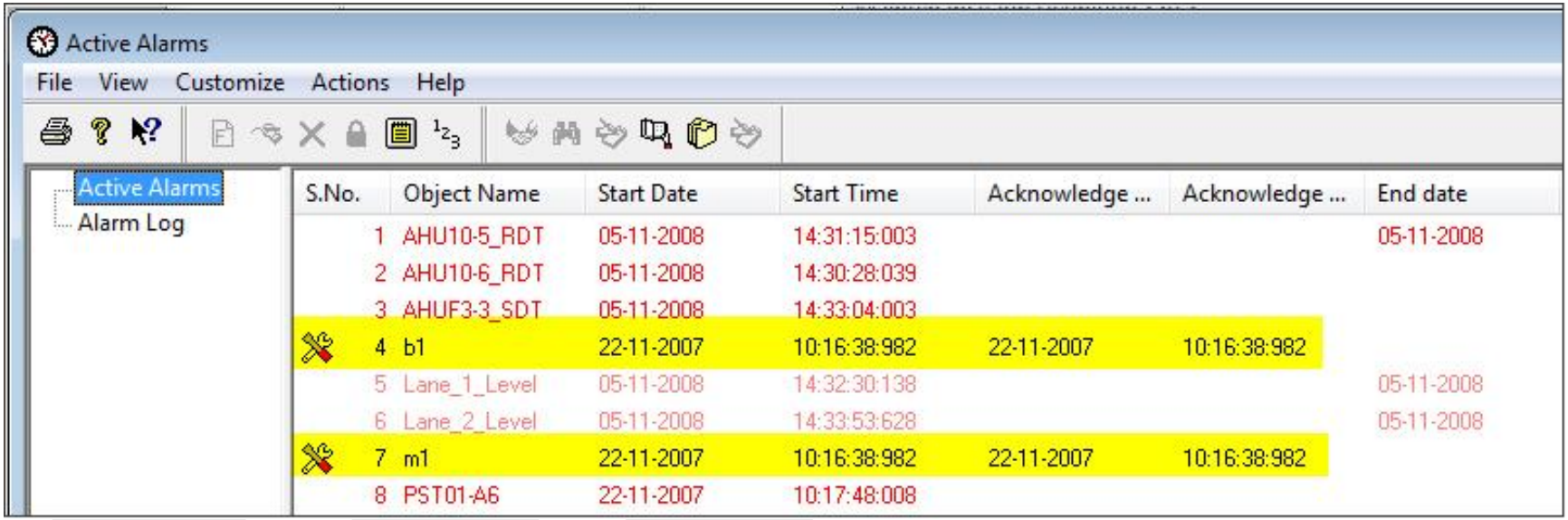
Handling maintenance alarms

- Visible both in Maintenance and Alarm List



Maintenance jobs in Alarm List

- Clearly marked with Maintenance icon
- Acknowledge in Alarm List or Maintenance List
- Complete from Maintenance List



Try it yourself ...



Exercise: Create Maintenance Jobs

Find presentations and exercises on www.7t.dk/igss



Exercise: Create Maintenance Jobs

Purpose

Learn how to create maintenance jobs for analog and digital objects. We will try three of the four maintenance types: periodical (**q1**), used time (**p1**) and changes (**p1**).

Duration

Max. 30 minutes.

Task 1: Create maintenance alarm texts in Definition

Before we create the maintenance jobs in the Maintenance program, we must create the alarm texts we need in Definition.

Step	Action
1.	In the **Definition** program, select **Edit → Alarm Texts**.
2.	Click **New** and create the following alarm text for the maintenance alarm on **q1**: • **Acknowledge color:** Light red, no blink. • **Alarm text:** "Calibrate flow sensor" • **Instructions:** "Use Calibrator AB 1001"
3.	Click **OK** to save the alarm text.
4.	Click **New** again and create the first maintenance alarm text for **p1**: • **Acknowledge color:** Light red, no blink. • **Alarm text:** "Lubricate pump bearings" • **Instructions:** "Use SuperLub for optimal lubrication" Click **OK** to save the alarm text.
5.	Click **New** again and create the second maintenance alarm text for **p1**: • **Acknowledge color:** Light red, no blink. • **Alarm text:** "Visually inspect rotating parts" • **Instructions:** "Make sure you're inspecting all rotating parts" Click **OK** to save the alarm text.



6. Install the configuration.

Task 2: Start Supervise and Maintenance

Before we can create the jobs, we need to make sure that **Maintenance** is running when the IGSS application is started.

Step	Action
1.	Open **System Configuration** and go to the tab **Startup** and under **Auto start** select **Maintenance**.
2.	Click **File → Save and Exit**.
3.	Click the IGSS Starter icon  IGSS32 Starter to start the system and the **Maintenance** module is started automatically..

Task 3: Create the maintenance job for q1

For the flow sensor we want to create a maintenance job that ensures that the flow sensor is calibrated every 60th day. Because we want to see the maintenance alarm right now, we create it as 2 minutes instead.

Step	Action
1.	In Maintenance, select **Actions → New Object** or click  . The **Select Object** dialogue appears.
2.	Under **Types**, select **Analog**.
3.	Under **Name**, select **q1**.
4.	Under **Maintenance type**, select **Periodical** and click **OK**. The **Maintenance Job Definition** dialogue appears.
5.	On the **Periodical** tab, enter 2 minutes as your **Service interval**.
6.	In the **Alarm number** drop-down list, select the “Calibrate flow sensor” alarm text created in Task 1.
7.	Click **Maintenance Info** to attach the operator instructions. The **Maintenance Information** dialogue appears.
8.	On the **Job Description** tab, type a filename for the maintenance job description (for example, “q1_cal_sensor.job”) and type these instructions: “When you calibrate the sensor, make sure you’re using Calibrator AB 1001.” Click **OK**.



9. Click **OK** again to save the job. The job now appears in the **All Maintenance Objects** list.

Task 4: Create the maintenance jobs for p1

For the pump we want to create two maintenance jobs: one where we ensure that the pump bearings are lubricated after 1000 hours of operation and one where we ensure that the rotating parts are visually inspected after 1000 state changes.

Step	Action
1.	In Maintenance, select **Actions → New Object** or click  . The **Select Object** dialogue appears.
2.	Under **Types**, select **Digital**.
3.	Under **Name**, select **p1**.
4.	Under **Maintenance type**, select **Used Time** and **Changes** and click **OK**. The **Maintenance Job Definition** dialogue appears.
5.	On the **Used Time** tab, enter 3 minutes (1000 hours in real life) as your service interval.
6.	In the **Alarm number** drop-down list, select the “Lubricate pump bearings” alarm text created in Task 1.
7.	Click **Select States** and move the **Slow**, **Medium** and **Fast** states to the **Monitor states** list. Click **OK**.
8.	Click **Maintenance Info** to attach the operator instructions. The **Maintenance Information** dialogue appears.
9.	On the **Job Description** tab, type a filename for the maintenance job description (for example, “p1_lub_bearings.job”) and type these instructions: “When you lubricate the bearings, make sure you’re using SuperLub.” Click **OK**.
10.	Click the **Changes** tab to create the second maintenance job.
11.	In the **Maintain after** box, type 5 changes as your service interval (1000 changes in real life).
12.	In the **Alarm number** drop-down list, select the “Visually inspect rotating parts” alarm text created in Task 1.



13. Click **Select States** and move all four states to the **Monitor states** list. Click **OK**.

14. Click **Maintenance Info** to attach the operator instructions. The **Maintenance Information** dialogue appears.

15. On the **Link** tab, we make a reference to the IGSS[32] Getting Started manual as an example of referring to the component manufacturer's documentation. Click **Browse** and find the **\Gss\Eng** subfolder of your installation path. Select the file **EngStart.pdf**. The operator will then be able to call up this manual when the maintenance alarm occurs.

16. Click **OK** to save the maintenance job and close the dialogue.

Task 5: View, acknowledge and complete maintenance alarms

In this last task, we'll handle the maintenance alarms that will result from the maintenance jobs we just created. Remember that the maintenance alarm will occur in two contexts: in the Alarm List (because we attached an alarm text) and in the Maintenance List.

Step	Action
1.	When the first maintenance alarm occurs in the Alarm List, acknowledge it.
2.	In Maintenance, select the alarm in the **Active Maintenance Objects** list and click the  icon in the toolbar to view the operator instructions. Click **OK** when you've read the instructions.
3.	Complete the alarm by clicking the  icon. The maintenance job will now be moved to the **All Maintenance Objects** list.
4.	Click **All Maintenance Objects** in the tree view to see the result.

YOU HAVE SUCCESSFULLY COMPLETED THE EXERCISE !!